Chain Of Product

Network and Computer Security

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Table of contents

01 Secure document format

02 Built infrastructure

03 Secure channels

04 Security challenge

05 Key distribution

06 Results and conclusion

Chain of Product Scenario

- Business setting of mutually distrusting companies, Transaction Server acts as a mediator.
- DvP transactions are JSON documents with sensitive business data.
- Transaction Server & Database are **trusted** but have no access to the content of the transactions.

```
{
  "id": "doc-001",
  "timestamp": 17663363400,
  "seller": "alice",
  "buyer": "bob",
  "product": "Indium",
  "units": 40000,
  "amount": 90000000
}
```

01

Secure document format

Secure Document – Format

DvP Transaction

```
{
  "id": "doc-001",
  "timestamp": 17663363400,
  "seller": "alice",
  "buyer": "bob",
  "product": "Indium",
  "units": 40000,
  "amount": 90000000
}
```

Secure Document Output:

```
{
  "sender":"alice",
  "receiver":"bob",
  "timestamp":"1.76633634E10",
  "seq_num":"1.0",
  "content_cipher":"diMgD1CEobOUBqKSXD..."
}
```

Document

```
c7yoJVFcDPl0hvF+xEtHBu6kU0t...
```

Transaction Key

```
SJldK/eWVgzi7GF32zQHcjBfIqB6...
```

Agreement Signature

Transactions and their Metadata

- The transactions are stored in the DB alongside their **metadata**.
- Access control and security requirements are enforced with the help of the metadata.

Metadata holds:

➔ **List of encrypted transaction keys** (sharedKeys[]): K_tx wrapped with RSA.

➔ **Agreement signatures**: seller/buyer signatures over the transaction.

➔ **Share logs**: list of who the transaction was shared with and by who.

02

Built infrastructure

System components

Client

Java (JAX-RS Client API)

Group Server

Java Jakarta EE (JAX-RS)

Transaction Server

Java Jakarta EE (JAX-RS)

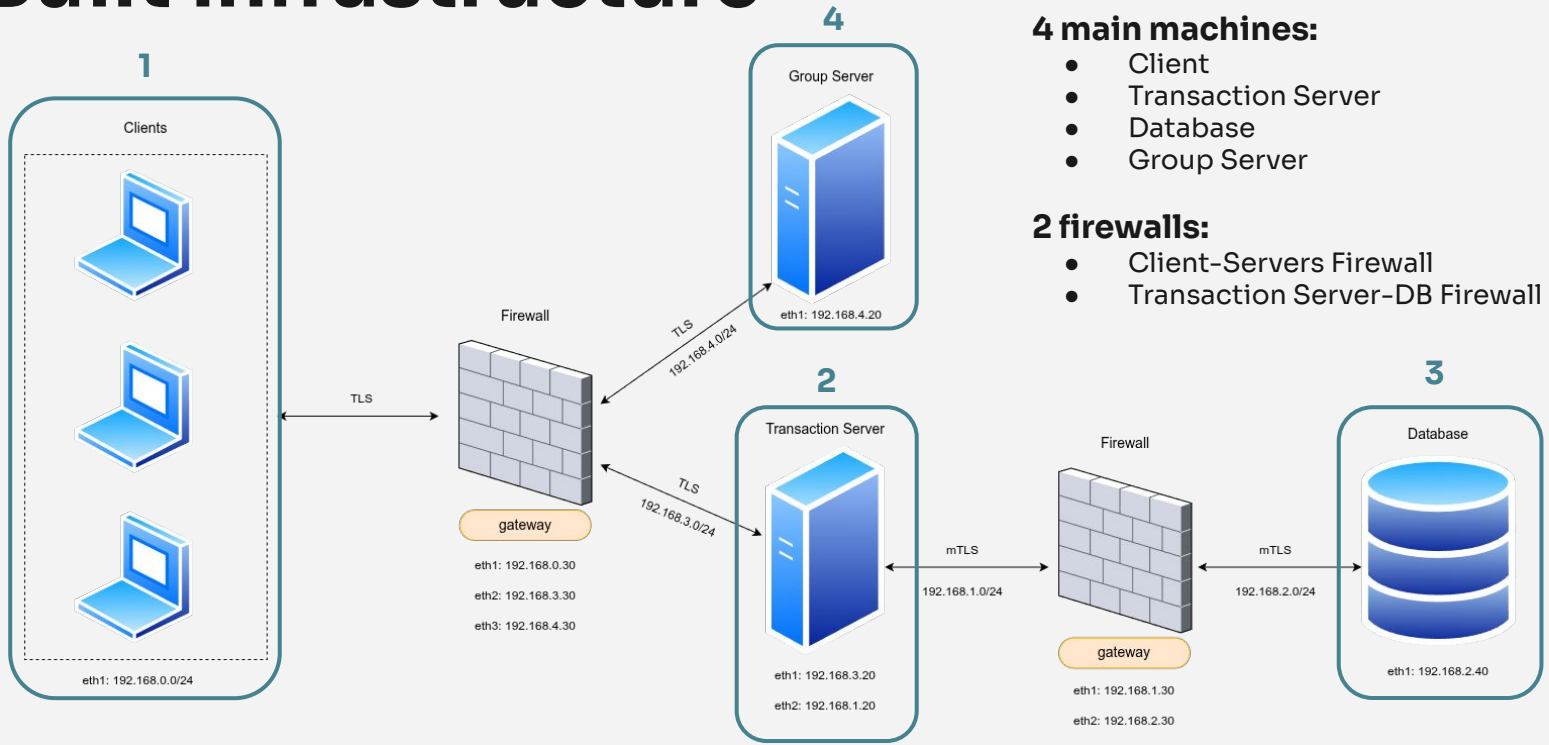
Database

PostgreSQL + Docker

Firewalls

iptables

Built infrastructure



4 main machines:

- Client
- Transaction Server
- Database
- Group Server

2 firewalls:

- Client-Servers Firewall
- Transaction Server-DB Firewall

Built infrastructure

Additional Security - IP Table rules

Transaction Server

- Address is unknown to clients
- Only accepts connections on **port 8443** from the Clients network.

Group Server

- Address is unknown to clients
- Only accepts connections on **port 8081** from the Clients network.

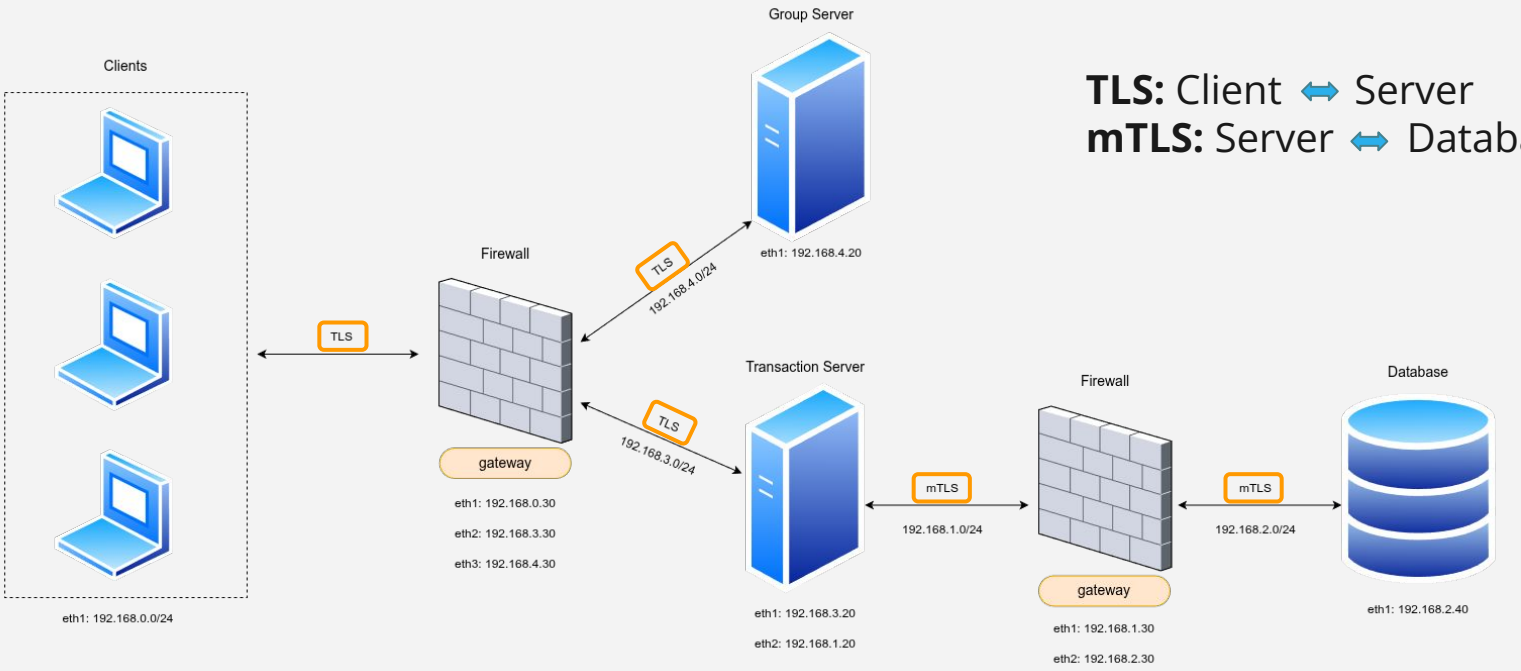
Database

- Only accepts connections on **port 5432** from the Transaction Server's IP address

03

Secure channels

Secure Channels



TLS: Client ⟺ Server
mTLS: Server ⟺ Database

04

Security challenge

Security challenge

Security Challenge A - Groups of partners

What changed

- New Group Server validates membership and controls transaction key access.
- The Group Server acts as a trusted gatekeeper
- The Transaction Server continues to handle individual disclosures and stores group-based shares, while delegating membership authorization to the Group Server

How it was implemented

- Group Server runs independently from the main Transaction Server
- The Group Server maintains a single key-pair to enable encryption/decryption of transaction keys shared with groups
- File-based storage that maps groups names to member lists (groups.json)

05

Key distribution

Key Distribution

User's Public Key

- Assumptions:

- Each user has access to every user's public key
- Each user assumes authenticity of pre-shared public keys

- RSA-2048 key pairs manually generated

Group Server's Key Pair

- Assumptions:

- Group Server is fully trusted by all users

- RSA-2048 key pair auto-generated at Group Server startup

- Group Server's public key retrieved dynamically via REST API

- Enables centralized access control through encryption

06

Results and conclusion

Results

Security requirements satisfied

- Confidentiality between buyer and seller is ensured via hybrid encryption
- Authentication on sharing is enforced through digital signatures
- Transaction integrity is maintained by using hashing
- Integrity of sharing lists is preserved through signed SharedLog entries

Security challenge implemented

- Group Server fully responsible for group management
- Isolated service, independent from the remaining system
- Parties are able to share transactions details with groups of partners

Conclusion

Main requirements and goals of the project were accomplished.

Possible future improvements:

- Public Key Infrastructure (PKI)
- Key revocation mechanism
- Web-based interface to improve usability